

Physics Before Einstein

A. Bork

Translated from the Russian version by V. A. Ugarov. Original English text: A. M. Bork, "Physics just before Einstein," Science 152 (No. 3722), 597 (1966).

Albert Einstein's 1905 paper dedicated to the special theory of relativity [1] is widely considered a concise exposition of revolutionary transformations in science. The purpose of this article is to outline certain aspects of the development of physics immediately prior to the creation of the special theory of relativity. A detailed enumeration of all the events occurring during this period would be neither engaging nor helpful in clarifying the issue, as these events are highly unequal in their historical significance. As often happens, a single direction of physics' development turned out to be dominant; it defined the physics of that era and raised genuinely interesting and important problems. Immediately before Einstein entered the scene, the activity of physicists was primarily focused on the leading physical theory of the time—electromagnetic theory. Late 19th-century physicists, of course, were not exclusively engaged in electromagnetic theory, but it was this theory that attracted universal attention. We shall focus our exposition on it, beginning with Maxwell's electromagnetic theory.

James Clerk Maxwell

Since it can hardly be said that Maxwell was an immediate predecessor of Einstein, I will dwell only briefly on his works. Virtually the entire content of Maxwell's theory is concentrated in three major papers [2] of 1856, 1861–1862, and 1864, a shorter paper of 1868, and finally, in the *Treatise on Electricity and Magnetism*, which appeared in 1873. The core ideas of the electromagnetic theory are already present in the second major paper, although they are presented in a clearer and more expanded form in the third paper and the *Treatise*.

Maxwell understood that the Faraday–Maxwell approach to electromagnetic phenomena differed from the continental European approach by its concept of the "field" (a word we first encounter in Maxwell's writings in 1865); this concept was the direct opposite of the concept of "action at a distance" (long-range action). The 1868 paper was specifically written to emphasize this distinction. Proponents of action-at-a-distance attempted to consider only the forces acting on bodies, such as gravitational forces. Proponents of the field concept also considered the space between the bodies—the "field." Although some physicists stubbornly refused to accept field ideas, usually for ontological reasons, believing that the field does not represent a physical reality, Maxwell's success in realizing the field viewpoint played a fundamentally important role in shaping the atmosphere of modern physics. Already since the 1930s, quantum field theory, which combines field concepts with quantization, has become the leading theory in physics.

We must clearly understand what Maxwell did for electromagnetic theory if we want to properly evaluate the contribution of his successors in this direction. We will consider the basic equations of Maxwell's theory in the form in which they were presented by Maxwell himself, as well as their consequences. Since the *Treatise* was much more widely known than Maxwell's other works, we will consider Maxwell's equations in the form in which they are presented there. This form of the equations coincides with what was presented in the 1864 paper. Maxwell used both components and quaternions in his equations (which were not used at all before the appearance of the *Treatise*). To facilitate the comparison between Maxwell's notation and modern notation, Figure 1 presents Maxwell's equations side-by-side with modern vector notation.

Anyone familiar with Maxwell's system of equations will immediately notice several differences. In his fundamental equations, Maxwell used not only field vectors, but also potentials. The concept of the vector potential was proposed by Maxwell to provide a mathematical description of the "electrotonic states" introduced by Faraday [4], with Maxwell noting, when explaining the introduction of the vector potential (contrary to what we think today), that the vector potential "may even be called the fundamental quantity of electromagnetic theory" ([3], vol. 2, section 540). The term "vector potential" itself appeared for the first time in the *Treatise*. Some of the Maxwell equations that are commonly written today, in particular the equations for $\text{div } \mathbf{B}$ and $\text{rot } \mathbf{E}$, are not explicitly written, although they are consequences of the equations that are listed. Maxwell clearly understood that he had written more equations than necessary ([3], vol. 2, section 615):

"These equations may be regarded as the principal relations among the quantities we are considering. They may be combined so as to eliminate some of these quantities, but our object at present is not to obtain the greatest compactness in mathematical formulae, but to establish all the relationships of which we have a clear conception. To eliminate a quantity which represents a useful idea would be a loss rather than a gain in this stage of our inquiry."

Almost the entire content of Maxwellian theory relates to the theory of optical phenomena, as presented in the second volume of the *Treatise* (chapters XX and XXI). After deriving the wave equation satisfied by the vector potential, Maxwell compares the experimental values of dielectric constants and refractive indices for various substances. Maxwell only briefly touches upon the question of energy (though he is firmly convinced that the field possesses energy) and barely touches upon the question of the electromagnetic field associated with an individual source. Our remarks should not be viewed as criticism of Maxwell; one could hardly expect from the founder of the theory a complete and exhaustive study of all its consequences.

Figure 1: Electromagnetic Field Equations

Below is the comparison of the equations presented in Maxwell's *Treatise on Electricity and Magnetism* (2nd edition) and modern vector notation:

Equations for	Maxwell's Notation	Modern Vector Notation
Magnetic Induction (A)	$\begin{aligned} a &= \frac{dH}{dy} - \frac{dG}{dz} \\ b &= \frac{dF}{dz} - \frac{dH}{dx} \\ c &= \frac{dG}{dx} - \frac{dF}{dy} \end{aligned}$	$\mathbf{B} = [\nabla \times \mathbf{A}]$
Electric Field Intensity (B)	$\begin{aligned} P &= c \frac{dy}{dt} - b \frac{dz}{dt} - \frac{dF}{dt} - \frac{d\psi}{dx} \\ Q &= a \frac{dz}{dt} - c \frac{dx}{dt} - \frac{dG}{dt} - \frac{d\psi}{dy} \\ R &= b \frac{dx}{dt} - a \frac{dy}{dt} - \frac{dH}{dt} - \frac{d\psi}{dz} \end{aligned}$	$\mathbf{E} = [\mathbf{v} \times \mathbf{B}] - \frac{\partial \mathbf{A}}{\partial t} - \nabla \phi$
Mechanical Force (C)	$X = cv - bw - e \frac{d\psi}{dx} - m \frac{d\Omega}{dx}$ (Only the first equation is shown)	$\mathbf{F} = [\mathbf{J}_T \times \mathbf{B}] + e\mathbf{E} - m\nabla\Omega$ (where $\mathbf{J}_T$ includes displacement current)
Magnetization (D)	$\begin{aligned} a &= \alpha + 4\pi A \\ b &= \beta + 4\pi B \\ c &= \gamma + 4\pi C \end{aligned}$	$\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$
Electric Current (E)	$\begin{aligned} 4\pi u &= \frac{d\gamma}{dy} - \frac{d\beta}{dz} \\ 4\pi v &= \frac{d\alpha}{dz} - \frac{d\gamma}{dx} \end{aligned}$	$[\nabla \times \mathbf{H}] = 4\pi\mathbf{J}_T$

Equations for	Maxwell's Notation	Modern Vector Notation
	$4\pi w = \frac{d\beta}{dx} - \frac{d\alpha}{dy}$	
Electric Induction (F)	$\mathfrak{D} = \frac{1}{4\pi} K \mathfrak{E}$ (Quaternion notation)	$\mathbf{D} = \frac{K}{4\pi} \mathbf{E}$
Conductivity (G)	$\mathfrak{K} = C \mathfrak{E}$ (Quaternion notation)	$\mathbf{J} = \sigma \mathbf{E}$
Total Current (H)	$u = p + \frac{df}{dt}$ $v = q + \frac{dg}{dt}$ $w = r + \frac{dh}{dt}$	$\mathbf{J}_T = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$
Free Charges (J)	$\rho = \frac{df}{dx} + \frac{dg}{dy} + \frac{dh}{dz}$	$\nabla \cdot \mathbf{D} = \rho$
Boundary Conditions (K)	$\sigma = lf + mg + nh + l'f' + m'g' + n'h'$	$(\mathbf{D}_2 - \mathbf{D}_1) \cdot \mathbf{n} = \sigma$
Magnetic Induction (L)	$\mathfrak{B} = \mu \mathfrak{H}$ (Quaternion notation)	$\mathbf{B} = \mu \mathbf{H}$

(Note in Figure 1: In the second equation of (A) in Maxwell's original text, there is a misprint; it should read dG/dz in the first term on the right and dF/dz in the second, as corrected in the vector notation.)

Maxwell's contemporaries did not initially accept his work. Tait and Kelvin (William Thomson) were close friends of Maxwell, but we have no evidence that Tait fully grasped the electromagnetic theory. Kelvin simply rejected this theory; he did not accept it even after Maxwell's death.

However, we can note with satisfaction that Maxwell lived to see the time when his electromagnetic theory began to gain traction. Figure 2 reproduces the first page of the abstract of Fitzgerald's paper, dedicated to the reflection and refraction of electromagnetic waves. The original of this abstract is kept in the library of Cambridge University. Maxwell mentions this matter in a letter to Stokes dated October 15, 1864 [5, p. 25], but he himself did not return to it. Maxwell was aware of Lorentz's work. Lorentz, in his doctoral dissertation completed in 1875, considered the reflection and refraction of light.

Those Who Generalized Maxwell's Theory

If we look closely at the activity that took place in the field of electromagnetic theory from Maxwell's death in 1879 until the appearance of Einstein's 1905 paper on the special theory of relativity, it is useful to draw a line between those who tried to modify or generalize the theory, and those who sought new results as consequences of the theory, i.e., developed it. Of course, it is difficult to draw such a distinction flawlessly, but it exists and reflects the actual state of affairs that we often encounter in theoretical physics. As is clear to us today, the greatest contribution to modern physics was made by those who developed Maxwell's theory. However, at the time we are discussing, this was by no means obvious, especially to those living in England. Nor can it be said that those who attempted to modify or generalize the theory always made negligible contributions to science. Let us look first at those who generalized the theory, and then return to those who tried to develop it.

Helmholtz and Boltzmann proposed their own version of Maxwellian electrodynamics; Boltzmann's book *Curry* [6], written on the basis of Boltzmann's lectures, introduced English readers to these ideas. Perhaps for continental Europe it is most correct to view this work as transitional from the "action-at-a-distance theory" associated with the names of Gauss, Weber, and Neumann, to the work of Hertz, which was already executed in the spirit of Maxwell's ideas.

Among the English physicists who developed Maxwell's ideas—here we are speaking mainly of Joseph Larmor and, to a lesser extent, J. J. Thomson, H. M. Macdonald, and Oliver Lodge—we can note a different line of thought. All the attention of this group was focused on the "ether"—the hypothetical medium in which electromagnetic energy was assumed to reside and through which electromagnetic waves propagated. Apparently, the peak of this activity was J. Larmor's book *Aether and Matter*, published in 1900 [7], which had a significant impact. Long before the publication of this book, Larmor wrote enthusiastically about the ether in a letter addressed to Oliver Heaviside (the letter is dated October 12, 1893) [8]:

"I feel that I have arrived at a complete understanding of the ether, no less complete than our understanding of matter. I began by applying MacCullagh's theory of light to the ether in its full scope; then I transferred all electrical phenomena to the ether, starting from the usual limitations or assumptions (regarding dissipative processes); finally, I utilized Lord Kelvin's vortex theory of matter. Only gravitation was left aside, and it serves as a very useful deus ex machina as an illustration of one intricate point."

Almost all of those discussed, regardless of where they fall in our classification, believed in the ether. Some spoke of the ether reservedly, while others viewed the question of the ether as the central, unresolved problem of electromagnetic theory. Even Maxwell himself, in his 1861 paper, proposed a model using rotational motion and guiding valves (something like ball bearings; see Fig. 3), though he subsequently abandoned it. Nevertheless, he never gave up the idea of finding a mechanical foundation for electromagnetic theory.

Belief in the existence of the ether is not at all connected with whether Maxwell's point of view on electricity and magnetism is accepted or not. The most striking example is Lord Kelvin, who was negative towards Maxwell's theory but spent his whole life, albeit unsuccessfully, trying to develop a concept of the ether.

At the end of his *Treatise* ([3], vol. 2, sections 865, 866), Maxwell returns to the ideas of "action at a distance," briefly recalls them, and writes the following:

"In the minds of these eminent men, there seems to have taken root a certain prejudice, or, if you will, an a priori objection against the hypothesis that there exists a medium in which the phenomena of light and heat propagation take place, as well as the transmission of electric forces at a distance."

But, continuing his reasoning, Maxwell emphasizes that even this group, since it generally assumes a finite propagation time, implicitly uses the concept of the ether:

"Consequently, all these theories lead to the concept of a medium in which propagation occurs. And if we accept the existence of such a medium as a hypothesis, then, I believe, it must occupy an important place in our research. We must attempt to construct a mental representation of all the details of its manifestations. This was the constant goal of my treatise."

The persistence with which the ether hypothesis was advocated in physics at the end of the 19th century has been noted repeatedly. It reflected a mechanical approach to physical reality, an approach that physicists wanted to preserve at all costs.

Vectors

Today it is quite difficult to understand why twenty years passed from the moment the theory of electromagnetism was formulated in the period from 1861 to 1865 until the first attempts were made to clarify all the consequences of the theory. Reading textbooks, one might think that as soon as a new theory appears, everyone rushes to test it in order to immediately decide whether to accept or reject it. But in fact, often no one is in a hurry. In the case of interest to us, the development of electromagnetic theory was delayed both because of the need to develop a new mathematical technique and because of the need to introduce a new system of notation—vectors.

Independently of each other, Heaviside and Gibbs isolated the vector aspects from Hamilton's quaternions and succeeded in building pure vector algebra and vector calculus; both had the specific intention of using them in electromagnetic theory; the works of these two authors differed essentially only in notation. Heaviside speaks of vectors in his book *Electromagnetic Theory* ([9], vol. I) as follows:

"My acquaintance with quaternions came about quite differently. Maxwell presented his results in the Treatise in quaternion form. I turned to Tait's book to learn how to deal with them. I had many difficulties, but in the end, I found that quaternions could be quite reasonably used when working with vectors. However, trying to apply quaternions in electromagnetic theory, I realized that this was very inconvenient. Quaternions in their vector aspect seemed antiphysical and unnatural, and did not agree at all with ordinary scalar mathematics. And I completely abandoned quaternions, began to use pure scalars and vectors, and starting from 1883 used a simple vector algebra in my works."

The choice of notation may seem far less important, because it affects only the form, but not the content of the theory. However, the history of science knows opposite examples where notation played a primary role. This was the case, for example, in the generalization of the special theory of relativity and the transition to the general theory of relativity. Most of the fundamental quantities in electromagnetic theory are vectors, so working exclusively with their components is simply tedious. Maxwell, under the influence of his friendship with Tait, used quaternion notations in many chapters of the *Treatise*, but rarely used them in derivations. But, as Gibbs and Heaviside understood, he actually needed only vectors of a certain type, so quaternions that were not vectors are almost never found in the *Treatise*.

Other English physicists, unlike Heaviside, refused to introduce vectors. Kelvin spoke out very sharply against vectors [10]. Jeans, in various editions of his *Treatise*, strictly used vector components. The Germans at first avoided vectors, but their appearance can first be detected in Lorentz's work in 1892. However, already Föppl in his 1894 textbook used Heaviside's notation, and Lorentz's article [11], dedicated to electromagnetic theory in the *Encyklopädie der mathematischen Wissenschaften* (1904), fully uses vector notations, differing only slightly from the notations of Gibbs and Heaviside. In general, it can be said that those who were interested in the development of electromagnetic theory were no less interested in vectors; this case can be used as a striking example of how proper notations significantly assist the development of physics.

Oliver Heaviside

Oliver Heaviside received an honorary degree in Göttingen in 1905; the corresponding citation stated that he was "probably the very first of Maxwell's followers." I am inclined to consider this assessment correct. At the same time, even specialists in the history of physics have almost completely forgotten about Heaviside's role. His name is remembered in connection with two innovations introduced into physics, which are of by no means primary significance—the rationalized system of units in electromagnetic theory and the Heaviside

layer of charged particles above the Earth's surface, as well as in connection with the technique of operational calculus. But even a superficial glance at his works on electricity [12, 13] or his book *Electromagnetic Theory* [9] (consisting of five volumes) shows that it is to Heaviside that the development of Maxwell's theory in a wide variety of directions belongs. Since Heaviside's works are now little known, I will focus on some of them.

Heaviside's education was highly unusual. He did not graduate from university and was self-taught. Nevertheless, he managed to quickly become a telegraph engineer thanks to the patronage of his uncle, Sir Charles Wheatstone. It is possible that it is because of this circumstance that Heaviside's first works are dedicated to the Wheatstone bridge. From Heaviside's notebooks, it can be established that he sent one of these works to Maxwell, and in Maxwell's papers, notes regarding this work were found (Fig. 4). Heaviside's works in electromagnetic theory open with a work completed in early 1880 and conclude with a major series of works (1885–1887) under the title "Electromagnetic Induction and its Propagation" [14]. This series opens with Maxwell's equations written in a form that the modern reader will easily recognize. Heaviside was apparently the first to consider the equation for $\text{rot } \mathbf{E}$ as a fundamental equation, although the verbal formulation of this equation in integral form is given in Maxwell's 1868 paper. Heaviside particularly emphasized the symmetry between the electric and magnetic fields ("duplex" method).

Based on energy considerations, Heaviside concluded that potentials could be eliminated from the fundamental equations. Slightly later, Hertz, who worked with field components, arrived at the same equations, and for over twenty years these equations were commonly called "Maxwell's equations in the Hertz–Heaviside form" (in his 1905 paper dedicated to the special theory of relativity, Einstein called them the Maxwell–Hertz equations). Heaviside was so struck by the symmetry between the electric and magnetic fields that he sometimes introduced magnetic "charges" and magnetic currents, although he clearly realized that there was no experimental basis for this.

Heaviside, along with Poynting and independently of him, expanded the energetic treatment, including in it what is now called the Poynting vector; thus, it became possible to speak not only of the energy storage in the electromagnetic field (as Maxwell did), but also of the flow of energy. In modern textbooks, just as in Heaviside's 1892 paper presented to the Royal Society [15], the energy relations obtained are practically identical.

Flipping through Heaviside's works, we are convinced that his main contribution to electromagnetic theory consisted in the development of this theory, especially in determining the fields produced by various configurations of moving charges. His magnificent mathematical technique, applied to the solution of such problems, was not always properly understood (Bromwich, as it seems to me, was the first mathematician to appreciate it; this is clear from his correspondence with Heaviside) and was one of the reasons that hindered the publication of his works. Not to mention the masterful use of the operational method, Heaviside applied Bessel and related functions. Heaviside was the first to introduce the impulse $[\delta]$ function. Many others, among whom Fitzgerald, Wiechert, and Lorentz should be mentioned, also solved many specific radiation problems. We should consider the solution of such specific problems as the first step in the theoretical development of Maxwell's theory.

Electron Theory

The next step in the development of electromagnetic theory ([16], ch. XI) was made in a somewhat different direction, prompted by new experimental results—the discovery of the electron. This refers to what is now commonly called Lorentz's electron theory. However, one should also mention the names of those who made

significant contributions to the development of this theory. These are J. J. Thomson, Heaviside, Morton, Searle, Abraham, and Poincaré.

The basic idea of the theory is now widely known. A moving charged sphere, whose charge is by definition equal to unity, creates an electromagnetic field around itself; this field in turn acts on the sphere with a certain force. At some point, the prospects seemed encouraging; it seemed that "mass" could be excluded from fundamental physical concepts, explaining its appearance by the electromagnetic effect of "self-action" of the charged sphere. This tempting idea could not be realized, and today Lorentz's electron theory is mainly of historical interest to us, although it is returned to from time to time in our day. However, classical electron theory has retained its significance to our time, since some of its ideas return in the theory of relativity, although in a somewhat modified form.

The electromagnetic field of a charged sphere possesses energy and thereby forces the sphere to behave as if it had mass; we thus arrive at a relationship between mass and energy. Abraham, in particular, in his works strongly emphasized the need to consider energy and momentum. But the correspondence between them turned out to be more rigid: electromagnetic mass increased with speed, and the law of increase coincided with the relativistic law of mass dependence on speed: $(1 - v^2/c^2)^{-1/2}$. I will dwell on the experimental data concerning the dependence of mass on speed later, when describing Kaufmann's experiments.

Another relativistic idea that arose partly within the framework of electron theory is the contraction of lengths due to their motion; this idea was stimulated by the Michelson-Morley experiment and the "contraction hypothesis." The usual assumption for a stationary charged sphere was that the charge was uniformly distributed either throughout the volume of the sphere or over its surface. The shape of the stationary charge was considered strictly spherical. However, for a moving sphere, one had to think quite differently. Lorentz noticed some advantages of the "Heaviside ellipsoid," which contracted in the spirit of Fitzgerald-Lorentz; this contraction was later called relativistic contraction. Jammer [16] noted a more general and highly interesting aspect of the theory—an aspect that has overtones in modern physics. Ether theorists loved to "explain" electromagnetic theory using ether endowed with mechanical properties, but electron theory turned everything upside down; the central concept of classical mechanics—mass—began to be "explained" through fields and charges. Thus, the center of gravity of the theory clearly began to shift towards fields, moving away from particles.

In 1875, almost nothing had been completed in the direction of developing electromagnetic theory beyond what was done by Maxwell himself. But as soon as the development of the theory began, things went remarkably fast, and all this work was almost completely completed by 1900. The progress achieved can be clearly understood from two articles by Lorentz written in 1904 for the *Encyklopädie...*; the first of them is dedicated to electromagnetic theory, the second to electron theory. The modern reader will find in these articles many results well known to him. From the bibliography to these two articles (Fig. 5), one can see that only a few physicists took part in the development of the theory.

Although I by no means exhaust the list of those who took part in the development of the theory of the electromagnetic field, we must nevertheless move on. Turning to experimental achievements, I will only briefly mention Hertz and others who were able to generate electromagnetic radiation predicted by Maxwell's theory, since these crucial works are widely known.

As the basis for the approach to the experimental situation, we will take the experiments described by Lorentz in his attempt to build a general theory, entitled "Electromagnetic phenomena in systems moving with velocities smaller than that of light" [17]. First, the Michelson-Morley experiment is presented, set up with the

aim of measuring the speed of the Earth's motion relative to the ether. This experiment was carried out earlier than is usually thought; Michelson's first attempt, inspired by Maxwell's letter in *Nature*, was undertaken in 1881 [18], and a significantly improved experiment, in which Michelson used a massive slab floating in mercury, was performed in 1887 [19]. Consequently, by the time Lorentz wrote his review, the riddle of the Michelson experiment had been hanging over physicists for almost a quarter of a century. Nevertheless, not a single one of the leading physicists before Einstein even had the thought of abandoning the concept of the ether. Even those whose work almost did not rely on the concept of the ether (such as Heaviside), firmly believed in the existence of the ether.

The contraction hypothesis was the only serious attempt to explain the result of the Michelson-Morley experiment. This hypothesis assumed a physical contraction of the Michelson slab in the direction of motion. Lorentz first proposed this idea in 1892 and developed it in subsequent works; previously, he pointed out errors in the initial analysis of the experiment proposed by Michelson. The role of Fitzgerald seems to me more doubtful. In his published works, contraction is mentioned only in two bibliographies dedicated to Larmor's book *Aether and Matter*, where he does not claim at all that the idea of "contraction" belongs to him. He developed some thoughts related to contraction around 1892, mentioning them in a conversation with Oliver Lodge; Lodge mentions this very briefly in several articles, but this is completely insufficient to confirm the conclusion that the views of Fitzgerald and Lorentz were completely similar. Subsequently, Fitzgerald also told Lorentz that he was very interested in this problem, and Lorentz refers to him in his 1895 work. Larmor does not mention Fitzgerald's contributions at all in the book *Aether and Matter* (1900), in which only Lorentz is cited; however, only a year later, in an obituary dedicated to Fitzgerald, he attributes priority to him [20].

Most of the other experiments mentioned by Lorentz in his 1904 work also concern the motion of the Earth relative to the ether. He briefly dwells on the attempt of Rayleigh and Brace (1902–1904) to find out whether motion relative to the ether can lead to the appearance of double refraction in a moving body. The Trouton-Noble experiment is described in much more detail. These authors concluded that a torque should arise, tending to turn a charged capacitor so that its plates stand parallel to the direction of the Earth's motion; however, they could not detect the presence of such a torque even with the help of very sensitive torsion balances. Lorentz does not mention Lodge's experiments on light aberration at all.

The experimental result that attracted Lorentz's greatest attention in his 1904 work was of a completely different character. Several of the last pages of Lorentz's work are dedicated to comparing the theoretical conclusions from his electron theory with the experimental results of Kaufmann [21], obtained at the very beginning of the 20th century. These experiments were considered very important in their time, but very little is known about them now, and even then only in the most general terms. As already mentioned above, they are directly related to the electron theory. Kaufmann, using radium chloride obtained by him from Curie, studied the deflection of electrons moving at speeds close to the speed of light in electric and magnetic fields. From his measurements, it was possible to determine the masses of the electrons.

In his first publications, Kaufmann compared the results obtained with Searle's calculations and concluded that since only a negligible part of the mass depends on speed, the main part of the mass by its origin is not electromagnetic. Abraham and Heaviside emphasized the difficulties of the theoretical interpretation of these results, and Kaufmann, in turn, improved his experimental technique. New, more delicate experiments led him to completely opposite conclusions: "*The mass of electrons arising in Becquerel rays depends on their speed; this dependence agrees well with Abraham's formula.*" Thus, as it now seemed, the entire mass of the electron could be considered electromagnetic, and thereby mass could be excluded from the set of fundamental quantities. In a letter to *Nature*, Heaviside aligns himself with Lorentz, who in his 1904 paper pointed out that the dependence of mass on speed proposed by him (in our day this dependence is called the relativistic

dependence of mass on speed) fully agrees with Kaufmann's experimental results. It can be said that these experiments generally confirmed Lorentz's electron theory.

In conclusion, I would like to dwell on another group of experiments that had an extremely great influence on the development of physics. Electromagnetic theory was the foundation on which theories explaining newly discovered phenomena of the time—different types of "rays": Becquerel rays, X-rays, and cathode rays—were based. Most of these discoveries were made within a few years, around 1895. Unexpectedly and very dramatically, the world surrounding man became immeasurably more complex. One can to some extent imagine the excited atmosphere that prevailed at that time by reviewing Fitzgerald's letters [22] to Heaviside. We present several excerpts from these letters:

"[June 8, 1896]. This Röntgen radiation is very mysterious. Since the beginning of January, I have been strongly trying to prove that they represent ultraviolet vibrations, and in my time suggested to Lodge to set up an experiment to detect the expected vibrations of intermediate frequency. I myself do not have the appropriate equipment to obtain such vibrations..."

"[July 10, 1896]. I have not seen the description of the Swedish experiments with cathode rays. I am quite skeptical about many of these experiments, as well as about the homogeneity of the applied fields and their isotropy... it is very amusing that Hertz and Lenard believed so much that X-rays and cathode rays are one and the same thing that they managed to miss the discovery of Röntgen rays..."

"[September 28, 1896]. The conference was very interesting. I stayed with Lodge along with Carey Foster, Glazebrook, J. J. Thomson, and Rücker; one might say we formed a subcommittee of Section A. I forgot to mention Lenard. In his style, we always called them X-rays and avoided Röntgen's name. Extensive discussions unfolded on what cathode rays represent. Lenard and Bjerknes inclined to the view that these rays represent the propagation of something not connected with matter; all English physicists adhered to the corpuscular hypothesis. The difficulty for the former was that these rays are deflected in a magnetic field, and for the latter, the ability of cathode rays to penetrate thin metal films. According to observations, they pass through metal without any changes, judging by their ability to deflect. This is exactly what could be expected in the propagation of waves, but this requires collisions without change in the direction of motion, so that the deflection does not depend on what gas is behind the aluminum window. I devised a mechanical model of this phenomenon, and Lenard admitted that from the standpoint of mechanics there are no objections against this model, although, on the other hand, not a single one of the German physicists could propose any theory based on the concept of the ether that would explain the deflection of cathode rays in a magnetic field..."

The discussion of newly discovered phenomena is the main theme of Fitzgerald's letters at this time, and we can easily detect the important role of electromagnetic theory in this correspondence. Only newly discovered rays led to a whole series of wonderful and unexpected discoveries.

We cannot but marvel at that frantic, continuous activity which marked the period preceding 1905. The familiar Newtonian world began to crack in all directions. It was a matter not only of individual details, but all the mechanical foundations of this world were shaking. The ground for a revolution in physics was fully prepared.

References

1. A contrary point of view was held by Edmund Whittaker: *A History of the Theories of Aether and Electricity, The Modern Theories 1900–1926*, Dover, New York, 1954. The important role of Einstein in the history of the development of the theory of relativity is defended by Holton: *On the Origins of the Special Theory of Relativity*, Amer. J. Phys. 28, 627 (1960).
2. W. D. Niven, Ed., *The Scientific Papers of James Clerk Maxwell*, Cambridge Univ. Press, Cambridge, 1890: vol. 1, pp. 155–229 ("On Faraday's Lines of Force"); vol. 1, pp. 451–513 ("On Physical Lines of Force"); vol. 1, pp. 526–597 ("A Dynamical Theory of the Electromagnetic Field"); vol. 2, pp. 125–143 ("On a Method of Making a Direct Comparison of Electrostatic with Electromagnetic Force; with a Note on the Electromagnetic Theory of Light").
3. J. C. Maxwell, *A Treatise on Electricity and Magnetism*, Oxford, Ed. 2, 1881.
4. L. P. Williams, *Michael Faraday*, Basic Books, New York, 1964.
5. J. Larmor, Ed., *Memoir and Scientific Correspondence of the Late Sir George Gabriel Stokes*, Cambridge Univ. Press, Cambridge, 1907, vol. 2.
6. C. E. Curry, *Theory of Electricity and Magnetism*, Macmillan, London, 1897.
7. J. Larmor, *Aether and Matter*, Cambridge Univ. Press, Cambridge, 1900.
8. The cited letter belongs to Heaviside's correspondence collection at the Institution of Electrical Engineers in London.
9. O. Heaviside, *Electromagnetic Theory*, Electrician Print, and Publ. Co., Ltd., 1893.
10. A sharp controversy over two alternative systems of notation took place in the pages of *Nature* in the 1890s. See A. M. Bork, *Vectors vs. Quaternions, the Letters in "Nature"*, Amer. J. Phys. (1967).
11. H. A. Lorentz, *Maxwells Elektromagnetische Theorie. Weiterbildung der Maxwellschen Theorie und "Elektrontheorie"*, Encyklopädie der Mathematischen Wissenschaften, Teubner, Leipzig, 1904, Bände 2, 13–14.
12. A. M. Bork, Amer. J. Phys. (1967); see also: O. Heaviside, *Electrical Papers*, Macmillan, London, 1892.
13. I express my gratitude to G. Joseph, whose articles and oral communications supplemented my knowledge of Heaviside. A very interesting literary work concerning scientists close in spirit to Heaviside is: N. Wiener's *The Tempter*, Random House, New York, 1959.
14. O. Heaviside, *Electrical Papers*, Macmillan, London, 1892 (the work appeared first in *The Electrician*, January 3, 1885).
15. —, Phil. Trans. Roy. Soc. London A183, 423 (1893).
16. M. Jammer, *Concepts of Mass in Classical and Modern Physics*, Harvard Univ. Press, Cambridge, 1961.
17. The paper is reproduced in the collection: A. Einstein, H. A. Lorentz, H. Weyl, H. Minkowski, *The Principle of Relativity*, Dover, New York, 1923.
18. A. A. Michelson, Amer. J. Sci., No 3, 22, 120 (1881).
19. — and E. W. Morley, *Aether*, Phil. Mag. 24, 449 (1887).
20. A. M. Bork, Isis (1967).
21. W. Kaufmann, Phys. Zs. 4, 54 (1903).
22. Fitzgerald's letters to Heaviside are also located at the Institution of Electrical Engineers in London. This is a fairly extensive correspondence, comprising about 60 letters.